

Total Practicals to cover

Static Routing	Next hop method, Self exit method
Defaulting Routing	Loopback address setup
Dynamic Routing	RIP, EIGRP, OSPF
Router Password	Password, Secret Password, console, aux, telnet
SSH	Use SSH to get the access of router
DHCP	Assign IP's to PC dynamically
Telnet	Get the access of Router
DNS	Set and use the domain for specific IP
SMTP	Create a Mail server
Backup and Restore	Backup of Runconfig and startconfig using TFTP
Password Recovery	Recover password of Router using TFTP
Redistribution	RIP-OSPF, RIP-EIGRP, EIGRP-OSPF
NAT	PAT-NAT, Dynamic NAT, Static NAT
Inter-VLAN Communication	ROS,SVI
Port Security	Shutdown, Protect, Restrict
VLAN	Creation of VLAN and assigning port to Specific VLAN
ACL	Number Standard, Number Extended, Named Standard, Named Extended

Revision Done

Static Routing	Easy - Revised - no need to revise
Defaulting Routing	easy
Dynamic Routing	easy
Router Password	easy
SSH	easy
DHCP	medium
Telnet	easy
DNS	easy
SMTP	easy
Backup and Restore	medium
Password Recovery	hard
Redistribution	OSPF-RIP, RIP-EIGRP, EIGRP-OSPF
NAT	static - NAT, dynamic - NAT
Inter-VLAN Communication	medium
Port Security	easy
VLAN	easy
ACL	hard
VTP	easy

Classful network:

Uses default subnet mask /8 /16 /24

Classless network

Uses custom network /25, 26/, 27/

Router> user execution mode

Router# privilege mode

Router(config)# global configuration mode

Router(config-if)# interface mode

Router(config-line)# line mode telnet, ssh, password

Router(config-router)# routing configuration

Change the host name

#conf t

config# hostname *mumbai* or ho *mumbai*

config# exit

Basic Commands

show startup-config

shows saved configuration

write saves

running config on startup config

copy running-config startup-config

saves running config on startup config

show clock

shows routers clock

show ip protocols

shows routing protocol details

show protocols

shows IP configuration of interface

show history

shows last 10 commands protocol

Show terminal

shows terminal setting details

show flash

shows flash settings

del flash

deletes config settings of router

erase startup-config

deletes config settings of router

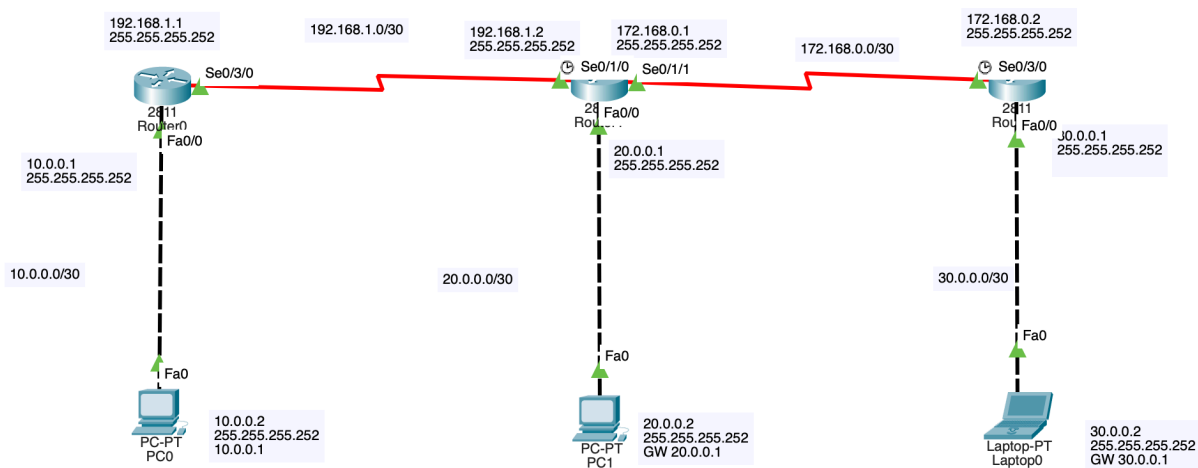
Static Routing

Notes

Static Routing

1. Next Hop

```
# conf t
config# ip route <unknown nw> <subnet mask> <next hop id>
```



```
R1(config)# ip route 20.0.0.0 255.255.255.252 192.168.1.2
R1(config)# ip route 172.168.0.0 255.255.255.252 192.168.1.2
R1(config)# ip route 30.0.0.0 255.255.255.252 192.168.1.2
```

```
R2(config)# ip route 10.0.0.0 255.255.255.252 192.168.1.1
R2(config)# ip route 30.0.0.0 255.255.255.252 172.168.0.2
```

```
R3(config)# ip route 20.0.0.0 255.255.255.252 172.168.0.1
R3(config)# ip route 192.168.1.0 255.255.255.252 172.168.0.1
R3(config)# ip route 10.0.0.0 255.255.255.252 172.168.0.1
```

2. Self Exit

```
# conf t
config# ip route <unknown nw> <subnet mask> <self serial interface >
```

```
R1(config)# ip route 20.0.0.0 255.255.255.252 se0/3/0
R1(config)# ip route 172.168.0.0 255.255.255.252 se0/3/0
R1(config)# ip route 30.0.0.0 255.255.255.252 se0/3/0
```

```
R2(config)# ip route 10.0.0.0 255.255.255.252 se0/1/0
R2(config)# ip route 30.0.0.0 255.255.255.252 se0/1/1
```

```
R3(config)# ip route 20.0.0.0 255.255.255.252 se0/3/0
R3(config)# ip route 192.168.1.0 255.255.255.252 se0/3/0
R3(config)# ip route 10.0.0.0 255.255.255.252 se0/3/0
```

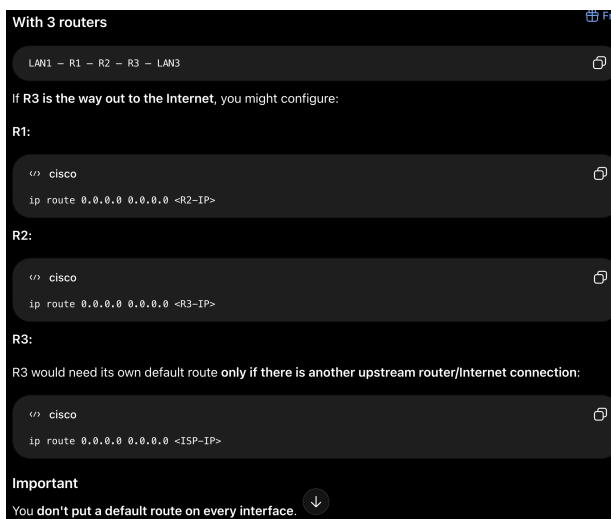
Default Routing

1.using next hop id:

```
config# ip route 0.0.0.0 0.0.0.0 <next hop id>
# do show ip route
```

2.using self exit:

```
config# ip route 0.0.0.0 0.0.0.0 s0/0
# do show ip route
```



Router Password

1. Normal Password

```
config# enable password nor@12  
config# no enable password
```

2. Secret/Encrypted

```
config# enable secret sec@123  
config# no enable secret
```

3. Console Password

1.To secure console mode

2.Must connect pc with router using console cable and to verify use terminal directly go to terminal and select default setting you will see the router>

```
config# line console 0  
config-line# password con@123  
config-line# login (must)  
config-line# no password
```

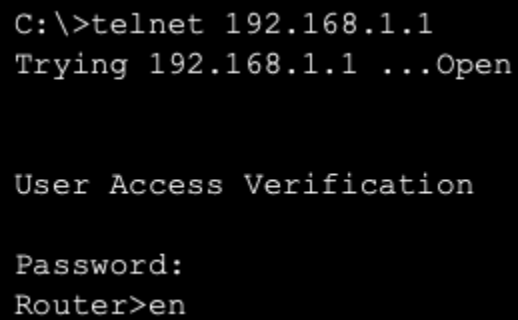
4. AUX

```
config# line aux  
config-line# password aux@123  
config-line# login
```

5. Telnet

1. Without setting this password you can't access the telnet

```
config# line vty 0 4
config-line# password tel@123
config-line# login
```



```
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

User Access Verification

Password:
Router>en
```

6. To encrypt all passwords

```
router(config)# service password-encryption
```

1. Just run this command on the start it will automatically encrypt all the passwords

SSH

Secure shell
Port no 21
Tcp
Application layer

On pc cmd
cmd# ssh -l *username connection-IP*

On router
config# ip domain-name corp
config# username *hemant* password 123456
config# crypto key generate rsa
config# line vty 0 4
config# transport input ssh
config# login local

```
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip domain
Router(config)#ip domain-n
Router(config)#ip domain-name corp.com
Router(config)#crypto key ge
Router(config)#crypto key generate rsa
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
Router(config)#ho r1
r1(config)#crypto key generate rsa
The name for the keys will be: r1.corp.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

r1(config)#line vty 0 4
*Mar 1 22:7:22.519: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 22:7:22.519: %SSH-5-ENABLED: SSH 1.5 has been enabled
r1(config-line)#transport input ssh
```

DHPC

Port no 67 and 68
UDP protocol

Router

```
config# hostname DHPC-router
config# ip dhcp pool corp
config# domain-name corp.com
config# network 192.168.1.1 255.255.255.0
config# default-router 192.168.1.1
Show ip dhcp binding
```

```
r1>en
r1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
r1(config)#int fa0/0
r1(config-if)#ip add 10.0.0.1 255.255.255.0
r1(config-if)#ip dhcp pool corp
r1(dhcp-config)#network 10.0.0.0 255.255.255.0
r1(dhcp-config)#default-router 10.0.0.1
r1(dhcp-config)#exit
r1(config)#
```

Go to IP configuration and convert all pc to static to dhcp

```
r1(config)#do show ip dhcp bi
r1(config)#do show ip dhcp binding
IP address      Client-ID/      Lease expiration      Type
                Hardware address
10.0.0.3        0000.0CB9.BB01  --
Automatic
10.0.0.4        00D0.BA0D.9425  --
Automatic
10.0.0.5        000C.85D2.5992  --
Automatic
r1(config)#
```

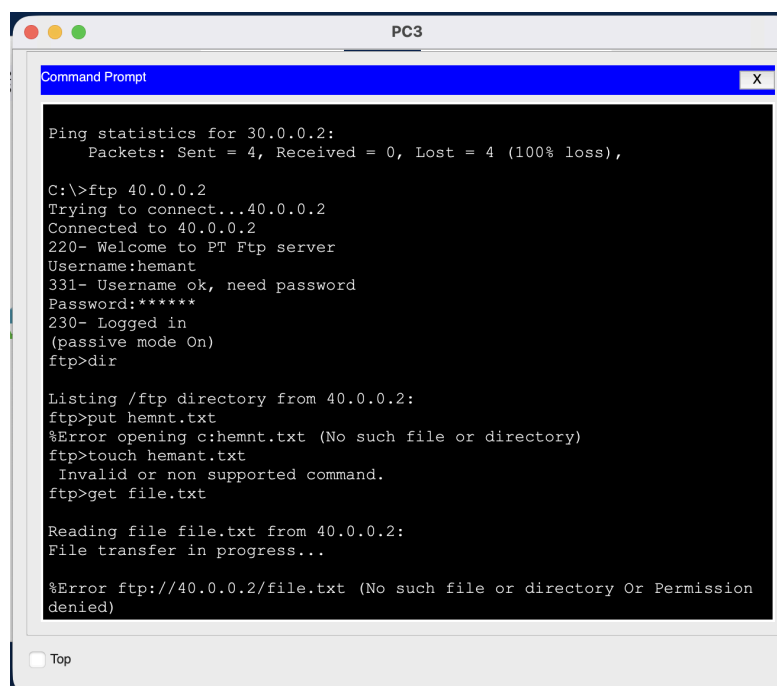
DNS

FTP

File transfer protocol

20,21 tcp

Application layer



```
Command Prompt
Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ftp 40.0.0.2
Trying to connect...40.0.0.2
Connected to 40.0.0.2
220- Welcome to PT Ftp server
Username:hemant
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 40.0.0.2:
ftp>put hemnt.txt
%Error opening c:\hemnt.txt (No such file or directory)
ftp>touch hemant.txt
Invalid or non supported command.
ftp>get file.txt

Reading file file.txt from 40.0.0.2:
File transfer in progress...

%Error ftp://40.0.0.2/file.txt (No such file or directory Or Permission
denied)
```

☐ Top

DNS server

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

FTP

Service

OnOff

User Setup

Username

Password

WriteReadDeleteRenameList

	Username	Password	Permission
1	cisco	cisco	RWDNL
2	hemant	123456	RWDNL

AddSaveRemove

File

POP3/SMTP

DNS server

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

EMAIL

SMTP Service

ONOFF

POP3 Service

ONOFF

Domain Name: corp.comSet

User Setup

User

Password

hemantniraj

+ - Change Password

Laptop0

PhysicalConfigDesktopProgrammingAttributes

Configure Mail

User Information

Your Name: niraj

Email Address: niraj@corp.com

Server Information

Incoming Mail Server 40.0.0.2

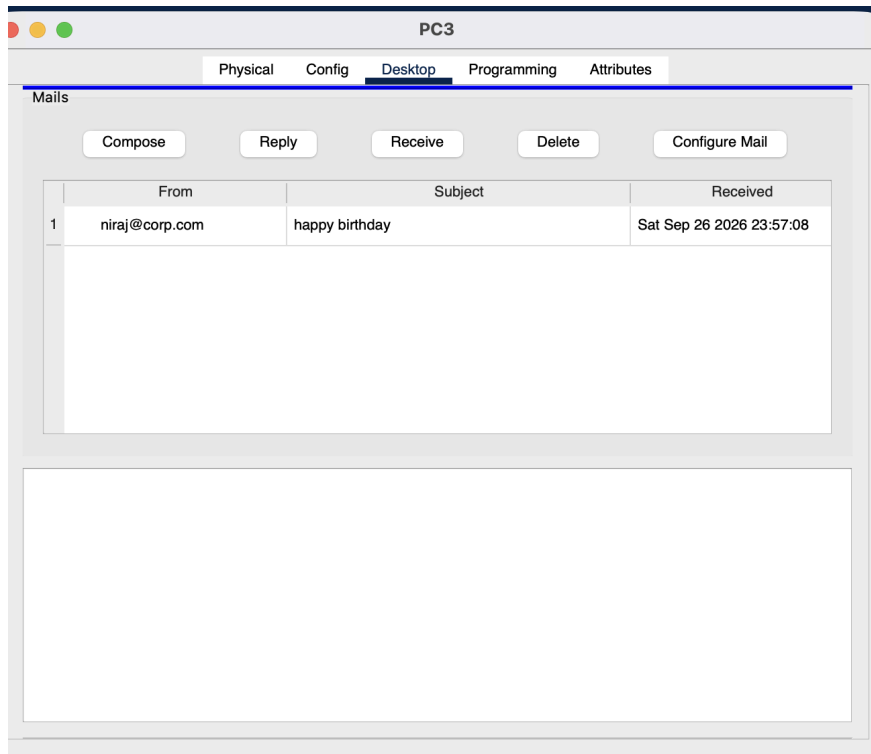
Outgoing Mail Server 40.0.0.2

Logon Information

User Name: niraj

Password:

Save Remove Clear Reset



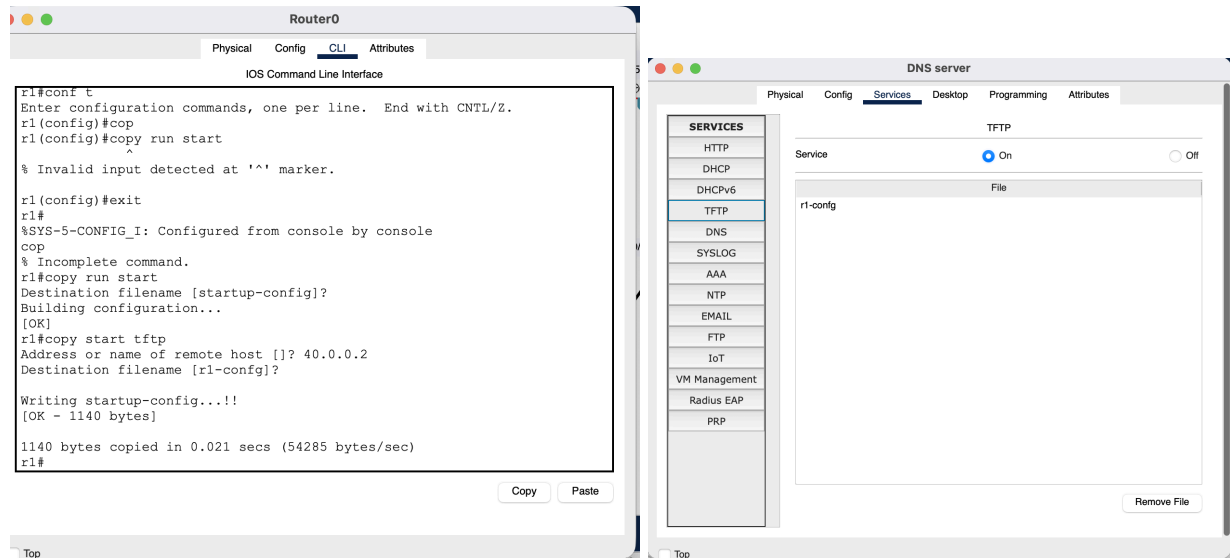
Backup and Restore

There are 2 types of backups

1. Startup-config backup & restore

It's a basic type of restore where we store the startup-config file on TFTP server and import it.

To backup
copy start tftp



To restore copy tftp backup

```

-----
rl#erase sta
rl#erase startup-config
Erasing the nvram filesystem will remove all configuration files!
Continue? [confirm]
[OK]
Erase of nvram: complete
%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
rl#show start
startup-config is not present
rl#copy tftp start
Address or name of remote host []? 40.0.0.2
Source filename []? r1-config
Destination filename [startup-config]?

Accessing tftp://40.0.0.2/r1-config...
Loading r1-config from 40.0.0.2: !
[OK - 1140 bytes]

1140 bytes copied in 0.016 secs (71250 bytes/sec)
rl#show start
Using 1140 bytes

```

2. IOS backup and Restore

Here we can do the password recovery

Show flash to get the name of run-config file .bin

```
System flash directory:
File Length Name/status
  3 33591768 2800nm-advispervicesk9-mz.151-4.M4.bin
  2 28282 sigdef-category.xml
  1 227537 sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

```
r1#copy flash tftp
Source filename []? 40.0.0.2
Address or name of remote host []?
r1#copy flash tftp
Source filename []? 2800nm-advipservicesk9-mz.151-4.M4.bin
Address or name of remote host []? 40.0.0.2
Destination filename [2800nm-advipservicesk9-mz.151-4.M4.bin]?
```

Now to **delete flash** use `del flash` and **reload** to restore

```

r1#del flash
Delete filename []?2800nm-advipservicesk9-mz.151-4.M4.bin
Delete flash:/2800nm-advipservicesk9-mz.151-4.M4.bin? [confirm]

r1#reload
Proceed with reload? [confirm]
System Bootstrap, Version 12.4(1r) [hqluong 1r], RELEASE SOFTWARE (fc1)
Copyright (c) 2005 by cisco Systems, Inc.

Initializing memory for ECC
..
C2800 processor with 524288 Kbytes of main memory
Main memory is configured to 64 bit mode with ECC enabled

Readonly ROMMON initialized

Boot process failed...

The system is unable to boot automatically. The BOOT
environment variable needs to be set to a bootable
image.
rommon 1 > |

```

Use command **tftpdnld** to setup tftp server

After that fill this all variables

The following variables are REQUIRED to be set for tftpdnld:

- IP_ADDRESS: The IP address for this unit
- IP_SUBNET_MASK: The subnet mask for this unit
- DEFAULT_GATEWAY: The default gateway for this unit
- TFTP_SERVER: The IP address of the server to fetch from
- TFTP_FILE: The filename to fetch

```

rommon 2 > IP_ADDRESS=192.168.1.1
rommon 3 > IP_SUBNET_MASK=255.255.255.0
rommon 4 > DEFAULT_GATEWAY=192.168.1.1
rommon 5 > TFTP_SERVER=40.0.0.2
rommon 6 > TFTP_FILE=
rommon 6 > TFTP_FILE=2800nm-advipservicesk9-mz.151-4.M4.bin
rommon 7 > tftpdnld

      IP_ADDRESS: 192.168.1.1
      IP_SUBNET_MASK: 255.255.255.0
      DEFAULT_GATEWAY: 192.168.1.1
      TFTP_SERVER: 40.0.0.2
      TFTP_FILE: 2800nm-advipservicesk9-mz.151-4.M4.bin
Invoke this command for disaster recovery only.
WARNING: all existing data in all partitions on flash will be lost!

Do you wish to continue? y/n: [n]: |

```

Do you wish to continue? y/n: [n]: y

.....[TIMED OUT]

TFTP: Operation terminated.

rommon 16 > TFTP_TIMEOUT=720000

rommon 17 > tftpdnld

IP_ADDRESS: 10.0.0.1

IP_SUBNET_MASK: 255.255.255.0

DEFAULT_GATEWAY: 10.0.0.1

TFTP_SERVER: 10.0.0.5

TFTP_FILE: 2800nm-advipservicesk9-mz.151-4.M4.bin

Invoke this command for disaster recovery only.

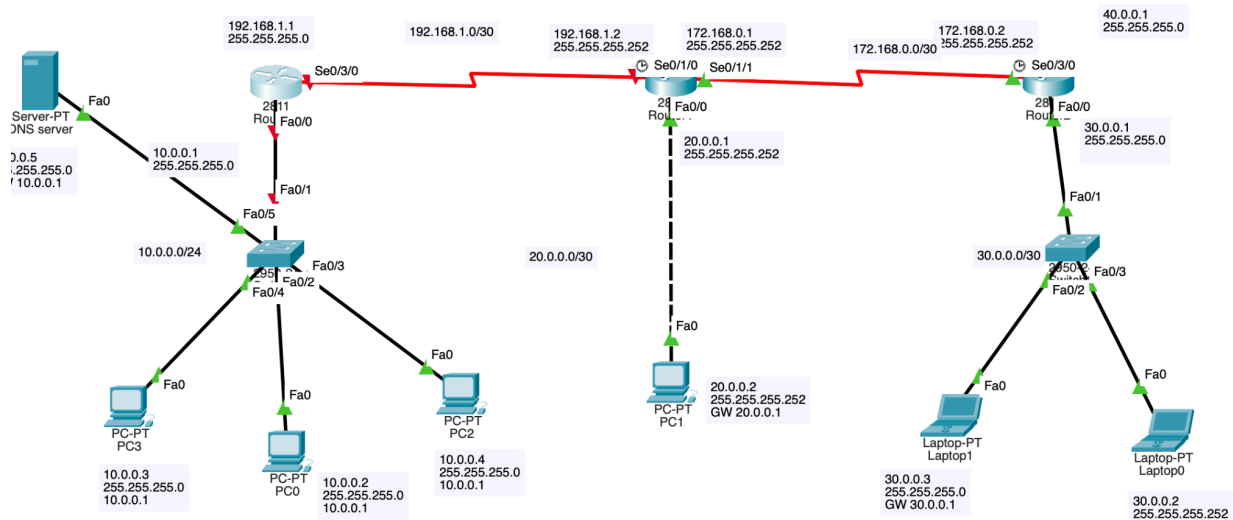
WARNING: all existing data in all partitions on flash will be lost!

Do you wish to continue? y/n: [n]: y

.....[TIMED OUT]

TFTP: Operation terminated.

rommon 18 >



3.Password Recovery

Set password and then copy the run file to the start we need start file

```
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#enable secret 123456
r3(config)#exit
r3#
```

```
r3#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
```

To enter in **rommon** mode turn off and on and quickly click ctrl + c in terminal

0x2142 is the recovery mode

Then go to recovery mood and then boot the file

```
rommon 1 > confreg 0x2142
rommon 2 > boot
System Bootstrap, Version 12.4(1r) [hqluong 1r], RELEASE SOFTWARE (fc1)
Copyright (c) 2005 by cisco Systems, Inc.
```

Copy the start file to run file and use no enable password to remove password and use **Confreg 0x2102 or config-register 0x2142** and then **reload** the it

```
Router>en
Router#copy start run
Destination filename [running-config]?

969 bytes copied in 0.416 secs (2329 bytes/sec)
r3#
%SYS-5-CONFIG_I: Configured from console by console
```

```
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#no enable secret
r3(config)#config-re
r3(config)#config-register 0x2102
r3(config)#do wr
Building configuration...
[OK]
r3(config)#do reload
```

Dynamic Routing

RIP

Router 1

Gateway of last resort is not set

```
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      10.0.0.0/24 is directly connected, FastEthernet0/0
L      10.0.0.1/32 is directly connected, FastEthernet0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/30 is directly connected, Serial0/3/1
L      192.168.1.1/32 is directly connected, Serial0/3/1
```

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#router rip

Router(config-router)#network 10.0.0.0

Router(config-router)#network 192.168.0.1

Router(config-router)#version

% Incomplete command.

Router(config-router)#version 2

Router(config-router)#

Router 2

Gateway of last resort is not set

```
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      20.0.0.0/30 is directly connected, FastEthernet0/0
L      20.0.0.1/32 is directly connected, FastEthernet0/0
      172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.168.0.0/30 is directly connected, Serial0/1/1
L      172.168.0.1/32 is directly connected, Serial0/1/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/30 is directly connected, Serial0/1/0
L      192.168.1.2/32 is directly connected, Serial0/1/0
```

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#router rip

Router(config-router)#network 20.0.0.0

Router(config-router)#network 172.168.0.0

Router(config-router)#network 192.168.1.0

Router(config-router)#version 2

Router(config-router)#

Router 3

```

    10.0.0.0/24 is subnetted, 1 subnets
S      10.0.0.0/24 [1/0] via 172.168.0.1
    20.0.0.0/30 is subnetted, 1 subnets
S      20.0.0.0/30 [1/0] via 172.168.0.1
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      30.0.0.0/24 is directly connected, FastEthernet0/0
L      30.0.0.1/32 is directly connected, FastEthernet0/0
    172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.168.0.0/30 is directly connected, Serial0/3/0
L      172.168.0.2/32 is directly connected, Serial0/3/0
    192.168.1.0/30 is subnetted, 1 subnets
S      192.168.1.0/30 [1/0] via 172.168.0.1

r3#
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#router rip
r3(config-router)#network 30.0.0.0
r3(config-router)#network 172.168.0.0
r3(config-router)#version 2
r3(config-router)#
```

To remove RIP just use **no router rip** at every router

And to verify if router setup is done correctly use

do show ip router rip

EIGRP

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/24 is directly connected, FastEthernet0/0
L    10.0.0.1/32 is directly connected, FastEthernet0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/30 is directly connected, Serial0/3/1
L    192.168.1.1/32 is directly connected, Serial0/3/1
```

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 10.0.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#no au
Router(config-router)#no auto-summary
Router(config-router)#
```

```
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/30 is directly connected, FastEthernet0/0
L    20.0.0.1/32 is directly connected, FastEthernet0/0
172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.168.0.0/30 is directly connected, Serial0/1/1
L    172.168.0.1/32 is directly connected, Serial0/1/1
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/30 is directly connected, Serial0/1/0
L    192.168.1.2/32 is directly connected, Serial0/1/0
```

```
Router#router eigrp 1
      ^
% Invalid input detected at '^' marker.
```

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 20.0.0.0
Router(config-router)#network 172.168.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 192.168.1.1 (Serial0/1/0) is up:
new adjacency
```

```
10.0.0.0/24 is subnetted, 1 subnets
S   10.0.0.0/24 [1/0] via 172.168.0.1
20.0.0.0/30 is subnetted, 1 subnets
S   20.0.0.0/30 [1/0] via 172.168.0.1
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   30.0.0.0/24 is directly connected, FastEthernet0/0
L   30.0.0.1/32 is directly connected, FastEthernet0/0
172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C   172.168.0.0/30 is directly connected, Serial0/3/0
L   172.168.0.2/32 is directly connected, Serial0/3/0
192.168.1.0/30 is subnetted, 1 subnets
S   192.168.1.0/30 [1/0] via 172.168.0.1

r3(config)#router eigrp 1
r3(config-router)#network 30.0.0.0
r3(config-router)#network 172.168.0.0
r3(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 172.168.0.1 (Serial0/3/0) is up:
new adjacency

r3(config-router)#no au
r3(config-router)#no auto-summary
```

Loopback Interface

Loopback Interface: is a logical interface

```
conf t
int lo0
ip add 5.0.0.1 255.0.0.0
```

```
exit
int lo1
ip add 9.0.0.1 255.0.0.0
```

```
exit
config#router eigrp 10
        network 5.0.0.0
        network 9.0.0.0
```

```
do sh ip route
```

```
goto pc5
```

```
ping 9.0.0.1
ping 5.0.0.1
```

OSPF

```
Router(config)#router ospf 1
Router(config-router)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter ar
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EG
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-I
inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```
        10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.0.0/24 is directly connected, FastEthernet0/0
L        10.0.0.1/32 is directly connected, FastEthernet0/0
        192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, Serial0/3/1
L        192.168.1.1/32 is directly connected, Serial0/3/1
```

```
Router(config-router)#network 10.0.0.0 0.0.0.255
% Incomplete command.
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
```

Gateway of last resort is not set

```
        20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/30 is directly connected, FastEthernet0/0
L        20.0.0.1/32 is directly connected, FastEthernet0/0
        172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.168.0.0/30 is directly connected, Serial0/1/1
L        172.168.0.1/32 is directly connected, Serial0/1/1
        192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, Serial0/1/0
L        192.168.1.2/32 is directly connected, Serial0/1/0
```

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 20.0.0.0 0.0.0.3 area 0
Router(config-router)#network 172.168.0.0 0.0.0.3 area 0
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#
00:48:05: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/1/0
LOADING to FULL, Loading Done

00:49:02: %OSPF-5-ADJCHG: Process 1, Nbr 172.168.0.2 on Serial0/1/1
LOADING to FULL, Loading Done
```

```
10.0.0.0/24 is subnetted, 1 subnets
S    10.0.0.0/24 [1/0] via 172.168.0.1
20.0.0.0/30 is subnetted, 1 subnets
S    20.0.0.0/30 [1/0] via 172.168.0.1
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/24 is directly connected, FastEthernet0/0
L    30.0.0.1/32 is directly connected, FastEthernet0/0
172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.168.0.0/30 is directly connected, Serial0/3/0
L    172.168.0.2/32 is directly connected, Serial0/3/0
192.168.1.0/30 is subnetted, 1 subnets
S    192.168.1.0/30 [1/0] via 172.168.0.1

r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#router ospf 1
r3(config-router)#network 172.168.0.0 0.0.0.3 area 0
r3(config-router)#network
00:48:56: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.2 on Serial0/3/0
LOADING to FULL, Loading Done

% Incomplete command.
r3(config-router)#network 30.0.0.0 0.0.0.3 area 0
```

Redistribution

RIP-EIGRP

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no au
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#
```

```
middle>en
middle#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
middle(config)#router rip
middle(config-router)#version 2
middle(config-router)#network 192.168.1.0
middle(config-router)#red
middle(config-router)#redistribute ei
middle(config-router)#redistribute eigrp ?
    <1-65535>  Autonomous system number
middle(config-router)#redistribute eigrp 1 ?
    metric  Metric for redistributed routes
    <cr>
middle(config-router)#redistribute eigrp 1 metric ?
    <0-16>      Default metric
    transparent Transparently redistribute metric
middle(config-router)#redistribute eigrp 1 metric 1 ?
    <cr>
middle(config-router)#redistribute eigrp 1 metric 1
middle(config-router)#exit
```

```
middle(config)#router eigrp 1
middle(config-router)#no a
middle(config-router)#no auto-summary
middle(config-router)#network 172.168.0.0
middle(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 172.168.0.2 (Serial0/1/1)
new adjacency
```

```
middle(config-router)#red
middle(config-router)#redistribute rip met
middle(config-router)#redistribute rip metric 1 1 1 1 1
middle(config-router)#
```

```
r3>en
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#router ospf 1
r3(config-router)#network 172.168.0.0 0.0.0.3 area 0
r3(config-router)#network 30.0.0.0 0.0.0.255 area 0
r3(config-router)#exit
r3(config)#no router ospf 1
r3(config)#router eigrp 1
r3(config-router)#network 172.168.0.0
r3(config-router)#network 30.0.0.0
r3(config-router)#no aut
r3(config-router)#no auto-summary
r3(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 172.168.0.1 (Serial0/3/0)
new adjacency
```

RIP-OSPF

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no au
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#
```

```
middle>en
middle#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
middle(config)#router rip
middle(config-router)#version 2
middle(config-router)#network 192.168.1.0
middle(config-router)#red
middle(config-router)#redistribute ospf 1 met
middle(config-router)#redistribute ospf 1 metric 3
middle(config-router)#exit
middle(config)#router ospf 1
middle(config-router)#network 172.168.0.0 0.0.0.3 me
middle(config-router)#network 172.168.0.0 0.0.0.3 area 0
middle(config-router)#redi
middle(config-router)#redistribute rip ?
    metric          Metric for redistributed routes
    metric-type     OSPF/IS-IS exterior metric type for redistributed routes
    subnets        Consider subnets for redistribution into OSPF
    tag             Set tag for routes redistributed into OSPF
    <cr>
middle(config-router)#redistribute rip subnets
middle(config-router)#
```

```
r3>en
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#router ospf 1
r3(config-router)#network 172.168.0.0 0.0.0.3 area 0
r3(config-router)#network 30.0.0.0 0.0.0.255 area 0
r3(config-router)#
```

EIGRP-OSPF

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no router rip
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#
04:44:15: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.2 on Serial0/3
LOADING to FULL, Loading Done
|

middle(config)#router ospf 1
middle(config-router)#network 192.168.1.0 0.0.0.3 area 0
middle(config-router)#red
middle(config-router)#redistribute eigrp metric ?
% Unrecognized command
middle(config-router)#redistribute eigrp 1 metric ?
<0-16777214> OSPF default metric
middle(config-router)#redistribute eigrp 1 metric 500 ?
metric-type OSPF/IS-IS exterior metric type for redistributed
subnets    Consider subnets for redistribution into OSPF
tag          Set tag for routes redistributed into OSPF
<cr>
middle(config-router)#redistribute eigrp 1 metric 500 subnets ?
tag Set tag for routes redistributed into OSPF
<cr>
middle(config-router)#redistribute eigrp 1 metric 500 subnets
middle(config-router)#exit
middle(config)#router eigrp
% Incomplete command.
middle(config)#router eigrp 1
middle(config-router)#network 172.168.0.0
middle(config-router)#
```

```

middle(config-router)#red
middle(config-router)#redistribute ospf 1 ?
    match   Redistribution of OSPF routes
    metric   Metric for redistributed routes
    <cr>
middle(config-router)#redistribute ospf 1 metric ?
    <1-4294967295>   Bandwidth metric in Kbits per second
middle(config-router)#redistribute ospf 1 metric 500 ?
    <0-4294967295>   EIGRP delay metric, in 10 microsecond units
middle(config-router)#redistribute ospf 1 metric 500 333 ?
    <0-255>   EIGRP reliability metric where 255 is 100% reliable
middle(config-router)#redistribute ospf 1 metric 500 333 255 ?
    <1-255>   EIGRP Effective bandwidth metric (Loading) where 255 i
loaded
middle(config-router)#redistribute ospf 1 metric 500 333 255 255
    <1-65535>   EIGRP MTU of the path
middle(config-router)#redistribute ospf 1 metric 500 333 255 255
    match   Redistribution of OSPF routes
    <cr>
middle(config-router)#redistribute ospf 1 metric 500 333 255 255
middle(config-router)#no auto
middle(config-router)#no auto-summary

```

```

r3>en
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#router ospf 1
r3(config-router)#network 172.168.0.0 0.0.0.3 area 0
r3(config-router)#network 30.0.0.0 0.0.0.255 area 0
r3(config-router)#exit
r3(config)#no router ospf 1
r3(config)#router eigrp 1
r3(config-router)#network 172.168.0.0
r3(config-router)#network 30.0.0.0
r3(config-router)#no aut
r3(config-router)#no auto-summary

```

Numbered Standard ACL

Access Control Lists

- Syntax for a Numbered Standard ACL:

```
access-list <ID#> <action> <source>
```

- Syntax for a Numbered Extended ACL:

```
access-list <ID#> <action> <protocol> <source> <destination>
```

Access Control Lists

- Syntax for a Numbered Standard ACL:

```
access-list <ID#> <action> <source>
```

- Syntax for a Numbered Extended ACL:

```
access-list <ID#> <action> <protocol> <source> <destination>
```

- <source> or <destination>

- Either are defined with: <IP Addresses> [Port]

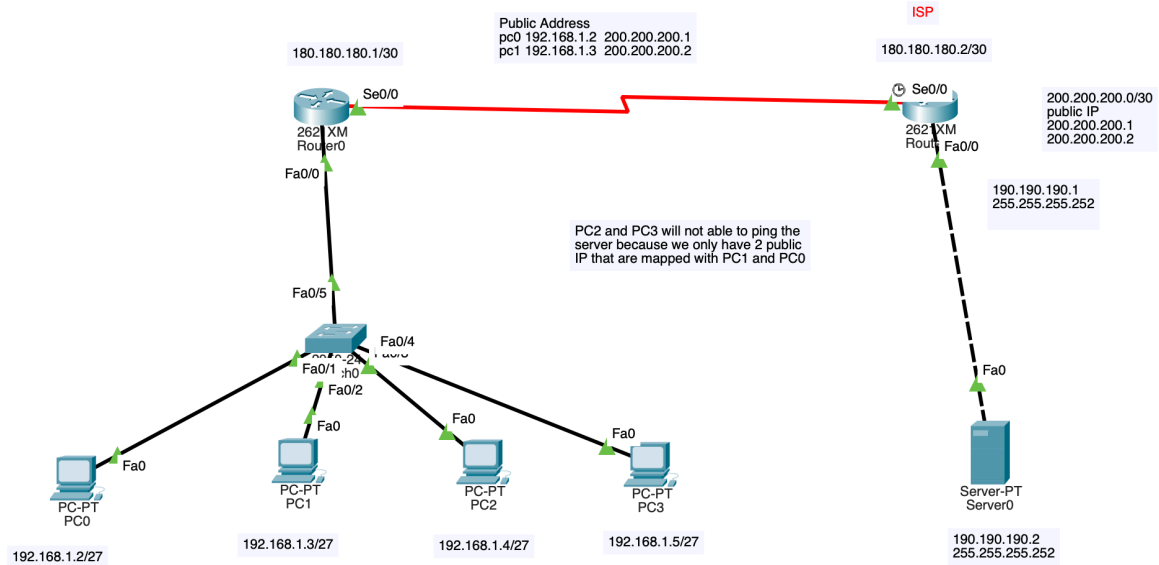
- To match a Subnet: <NetID> <Wildcard Mask>
- To match a single IP address: host <#.#.#.#>
- To match all IP addresses: any

Extended ACL

**Named
Standard ACL**

Extended ACL

Static-NAT



ISP router which is connected to internet

```
en
conf t
hostname ISP
int se0/0
ip address 180.180.180.2 255.255.255.252
clock rate 64000
no shutdown
exit
int fa0/0
ip address 190.190.190.1 255.255.255.252
no shutdown
exit
```

```
ip route 200.200.200.0 255.255.255.252 se0/0
```

On NAT router

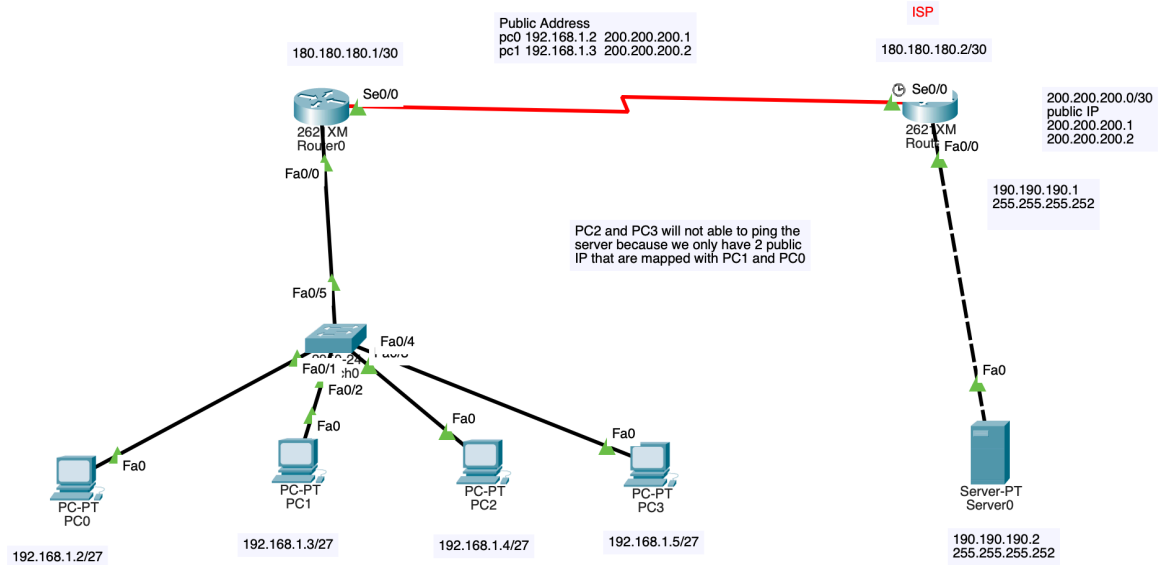
Step 1: Router0 (NAT router) interfaces

```
en
conf t
hostname NAT-Router
int fa0/0
 ip address 192.168.1.1 255.255.255.224
 no shutdown
 exit
int se0/0
 ip address 180.180.180.1 255.255.255.252
 no shutdown
 exit
```

```
int fa0/0
 ip nat inside
 exit
int se0/0
 ip nat outside
 exit
ip nat inside source static 192.168.1.2 200.200.200.1
ip nat inside source static 192.168.1.3 200.200.200.2
```

```
show ip int brief
show ip route
show ip nat translations
show ip nat statistics
```

Dynamic NAT



Router0 (NAT router)

```
en
conf t
hostname NAT-Router
int fa0/0
ip address 192.168.1.1 255.255.255.224
ip nat inside
no shutdown
exit
int se0/0
ip address 180.180.180.1 255.255.255.252
ip nat outside
no shutdown
exit
ip route 0.0.0.0 0.0.0.0 se0/0
```

Router1 (ISP)

```
en
conf t
hostname ISP
int se0/0
 ip address 180.180.180.2 255.255.255.252
 clock rate 64000
 no shutdown
 exit
int fa0/0
 ip address 190.190.190.1 255.255.255.252
 no shutdown
 exit
```



On Router0

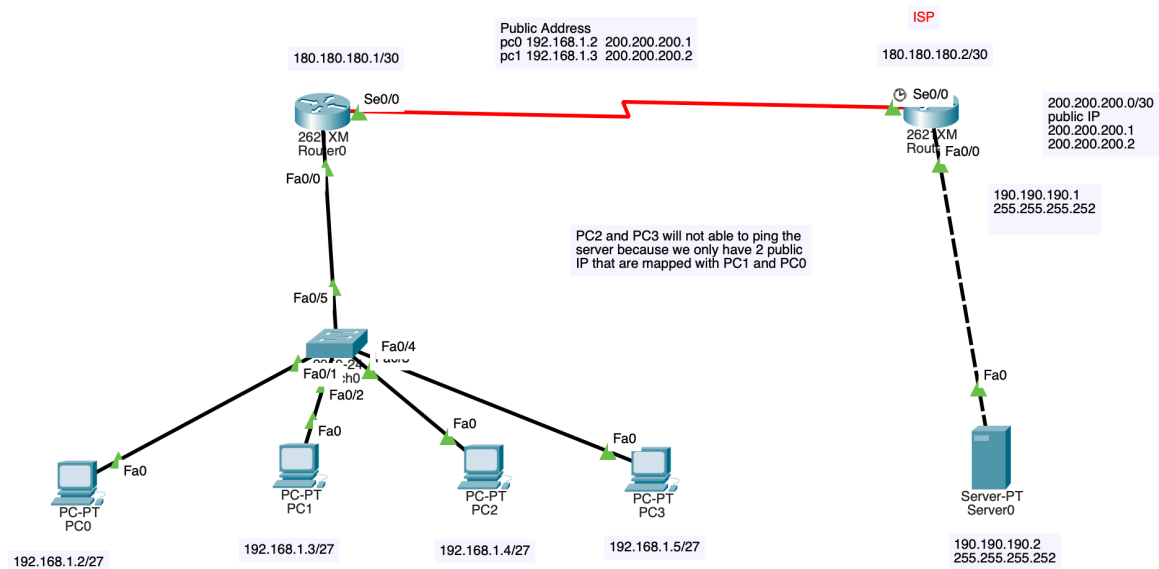
```
conf t
access-list 1 permit 192.168.1.0 0.0.0.31
ip nat pool MYP00L 200.200.200.1 200.200.200.2 netmask 255.255.255.252
ip nat inside source list 1 pool MYP00L
end
```

On ISP, a return route to the public pool:

```
conf t
ip route 200.200.200.0 255.255.255.252 se0/0
end
```



NAT-PAT



On Router0

```
access-list 1 permit 192.168.1.0 0.0.0.31
ip nat pool MYP00L 200.200.200.1 200.200.200.2 netmask 255.255.255.252
ip nat inside source list 1 pool MYP00L overload
end
```

On ISP (same as before):

```
ip route 200.200.200.0 255.255.255.252 se0/0
```

Option B: PAT with the interface IP (no pool, most common in real life)

On Router0

```
access-list 1 permit 192.168.1.0 0.0.0.31
ip nat inside source list 1 interface se0/0 overload
end
```

